Environment File

Indian Space Research Organisation

Team 15

December 7, 2025

1 Software Environment and Dependencies

This project was developed and verified using the following software tools, hardware description environments, and process design resources. The full workflow spans RTL design, simulation, synthesis, and physical design, along with MATLAB-based behavioral modeling and verification of different types modulator and filters.

1.1 Operating System

- **Primary OS:** Windows 11
- **Remote Access:** MobaXterm used for connecting to Linux-based Synopsys environment

1.2 HDL Design and Simulation

- **Vivado 2023.2**: Used for RTL design, SystemVerilog simulation, waveform verification, and behavioral testbench execution.
- **HDL Language:** SystemVerilog for RTL and testbench development.

1.3 Synthesis and Physical Design Tools

- **Synopsys Design Compiler (DC), Version X-2025.06**: Utilized for RTL synthesis, timing optimization, and generation of gate-level netlists.
- **Synopsys ICC2**: Applied for floorplanning, placement, routing, and full physical design (PnR) implementation.

1.4 Process Design Kit (PDK)

- **PDK: SCL180 CMOS Process** Technology libraries, timing files, and physical abstracts used during synthesis and physical design.

1.5 MATLAB Simulation Environment

The MATLAB environment was used for high-level architectural modeling, noise analysis, and exploration of Sigma-Delta ADC performance.

- **MATLAB Version:** R2025a/R2025b
- **Purpose:**
  - Ideal and non-ideal Delta Sigma ADC architecture modeling
  - Noise-injected simulations (flicker, thermal, jitter)
  - Digital decimation filter design and verification
  - ENOB vs. sampling-rate analysis
- **Execution Notes:** Simulations involve complex noise integration and require approximately 120 seconds of run-time on a high-performance CPU.

1.6 System Requirements

Processor: Intel Core i7 (or equivalent high-performance CPU).

1.7 MATLAB Toolbox Dependencies

Ensure the following toolboxes are installed for complete functionality:

- Mixed-Signal Blockset
- DSP System Toolbox
- Sigma-Delta Toolbox
- Filter Design Toolbox